IIA WEB DESIGN • UNIT 06

HTML Forms

Creating useful, accessible forms for information services

Bachelor of Library and Information Studies
Bachelor of Records Management Systems

LABEL
Book title

INPUT
Contact email

CHOICE
Collection point

BUTTON
Submit request

Learning outcomes

Form fundamentals for two professional settings.

Explain

Describe an HTML form and identify common control elements.

Build

Add labelled text inputs, choices, text areas and buttons.

Apply

Create a small fictional library or public records service form.

TAKEAWAY **A form is a user interface, not an entire information system.**

Where forms are used

Forms connect people with information services.

Library

Book requests, enquiries, registration for public events and feedback.

Records centre

Public record enquiries, assistance requests and access guidance.

Good design

Ask only what the visitor needs to complete a clear task.

TAKEAWAY **Good forms begin with the user’s purpose.**

What an HTML form does

Controls collect values entered or selected by a visitor.

Visible interface

Fields, options and buttons appear on a webpage.

Submitted values

A browser can send name–value pairs to a processing destination.

Processing

A separate service is needed to save or act on the data.

TAKEAWAY HTML alone cannot create a database or process a real request.

The form element

The outer container groups related input controls.

form

Contains the controls that belong to one request.

input

Represents a field for a value such as a book title.

button

Allows the visitor to submit the form.

HTML EXAMPLE

```
<form action="/requests" method="post">
  <label for="book">Book title</label>
  <input id="book" name="book"
         type="text" required>
  <button type="submit">Send</button>
</form>
```

TAKEAWAY **The form element surrounds the controls to be submitted.**

Action and method

A real request needs a destination that can process it.

action

The URL that receives the submitted information.

method

How data is sent; common values are get and post.

No endpoint?

A static HTML page does not create the service at /requests.

HTML EXAMPLE

```
<form action="/requests" method="post">
  ...
</form>

<!-- /requests is an example
     endpoint, not an actual
     running service. -->
```

TAKEAWAY **A form can be displayed without having a working back end.**

How a label helps

A clear label tells the visitor what belongs in a field.

Understand

“Book title” is more helpful than an unlabeled empty box.

Operate

Clicking an associated label focuses its input.

Assistive technology

Programs that read pages can identify named controls.

TAKEAWAY **Every important form field needs a meaningful label.**

Matching for and id

The value of for connects a label with the right control.

for="ref"

Points to the control whose id is ref.

id="ref"

Identifies this specific element on the page.

Unique IDs

Avoid using the same id for multiple elements.

HTML EXAMPLE

```
<label for="ref">Reference code</label>
<input id="ref"
       name="reference"
       type="text">
```

TAKEAWAY **The label's for must match the input's id.**

The name attribute

The name identifies a submitted value.

id

Used for labels and other page references.

name

Becomes a key in the submitted form data.

value

The content that belongs to that key.

HTML EXAMPLE

```
<label for="title">Book title</label>
<input id="title"
       name="book_title"
       type="text">

<!-- Example submitted key:
     book_title -->
```

TAKEAWAY **Use meaningful names for data fields.**

Text inputs

Use short text inputs for a title, reference or name.

type="text"

General short text such as a book title.

placeholder

A helpful hint, not the field label.

required

Useful when an empty value is unacceptable.

HTML EXAMPLE

```
<label for="book">Book title</label>
<input id="book" name="book"
       type="text"
       placeholder="e.g. Digital Libraries"
       required>
```

TAKEAWAY **Labels remain visible when the user starts typing.**

Email and telephone inputs

Choose a type that matches the information.

email

Checks the basic structure of an email address.

tel

Suitable for numbers with + or leading zeroes.

Not verification

Correct format does not prove ownership or identity.

HTML EXAMPLE

```
<label for="mail">Email</label>
<input id="mail" name="email"
       type="email">

<label for="phone">Telephone</label>
<input id="phone" name="phone"
       type="tel">
```

TAKEAWAY **A phone number is an identifier, not a quantity.**

Choosing correct input types

Different data calls for different controls.

text / search

Titles, keywords and public reference codes.

email / tel

Contact values with appropriate mobile keyboards.

number / date

Numeric quantities and visit dates where appropriate.

TAKEAWAY **The right control helps users enter correct information.**

Dates and numbers

Date and numeric fields support useful constraints.

date

A visit or consultation date.

number

A quantity, rather than a phone number.

min and max

Express a lower and upper numeric boundary.

HTML EXAMPLE

```
<label for="visit">Visit date</label>
<input id="visit" name="visit"
       type="date">

<label for="count">Copies</label>
<input id="count" name="count"
       type="number" min="1" max="3">
```

TAKEAWAY **Use numeric validation only for actual quantities.**

Multi-line textarea

A longer message should be easy to read and edit.

textarea

Accepts multiple lines of user-entered text.

rows

Suggests how many lines are visible.

Describe the task

Tell the user what kind of enquiry to write.

HTML EXAMPLE

```
<label for="message">
  Enquiry details
</label>
<textarea id="message"
          name="message"
          rows="4"></textarea>
```

TAKEAWAY **Use a textarea for comments rather than a single-line input.**

Single-choice radio buttons

Radio buttons are appropriate when exactly one option is allowed.

Same name

All options form one exclusive group.

Different values

Each option has its own submitted value.

Checked

Can set an initial choice where appropriate.

HTML EXAMPLE

```
<label><input type="radio"
  name="point" value="main">
  Main library</label>

<label><input type="radio"
  name="point" value="branch">
  Branch library</label>
```

TAKEAWAY **The shared name groups the options together.**

Independent checkboxes

A checkbox represents a separate true/false choice.

Independent

One checkbox does not switch off another.

Useful example

An optional request to receive service updates.

Careful wording

Agreements should be clear and not misleading.

HTML EXAMPLE

```
<label>
  <input type="checkbox"
         name="updates">
  Send general service updates
</label>
```

TAKEAWAY **Radio buttons select one; checkboxes can select more than one.**

Dropdowns with select

A dropdown can hold a predefined list of categories.

select

The input control containing a list.

option

One available choice in the list.

Empty first option

With required, invites a deliberate choice.

HTML EXAMPLE

```
<label for="cat">Category</label>
<select id="cat" name="category"
        required>
  <option value="">Choose one</option>
  <option value="report">Reports</option>
  <option value="guide">Guides</option>
</select>
```

TAKEAWAY **Use a dropdown when the available categories are known.**

Fieldset and legend

Group related choices under one clear question.

fieldset

A container for a related control group.

legend

The group heading or question.

Benefit

The user knows how the choices relate.

HTML EXAMPLE

```
<fieldset>
  <legend>Preferred contact</legend>
  <label><input type="radio"
    name="contact" value="email">
    Email</label>
  <label><input type="radio"
    name="contact" value="phone">
    Phone</label>
</fieldset>
```

TAKEAWAY **Use a legend to explain a group of radio buttons.**

The required attribute

A browser can detect common incomplete entries.

required

Requests a value before normal submission.

Try it

Leave the title blank and submit a practice form.

Limit

A non-empty value might still be wrong or unauthorised.

HTML EXAMPLE

```
<label for="title">Book title</label>
<input id="title" name="title"
       type="text" required>

<button type="submit">
  Send request
</button>
```

TAKEAWAY **A required field still needs clear instructions.**

Other useful constraints

Use simple limits that match the task.

minlength

Minimum text length for suitable inputs.

maxlength

Maximum character count for suitable inputs.

min / max

Bound a numeric quantity or relevant date range.

HTML EXAMPLE

```
<label for="book">Book title</label>
<input id="book" name="book"
       type="text"
       required minlength="3"
       maxlength="80">
```

TAKEAWAY **Constraints help prevent accidental mistakes.**

What browser checks cannot do

Client-side validation is helpful but limited.

Existence

It cannot establish that a particular book or record exists.

Identity

It cannot confirm the owner of an email address.

Permission

It cannot decide whether a visitor is authorised.

TAKEAWAY **Real services also need secure server-side checks.**

Buttons and their types

Buttons should make their effect clear.

submit

Sends the form to its destination after validation.

button

A non-submitting action, often controlled by JavaScript.

reset

Restores defaults, but may erase useful user work.

HTML EXAMPLE

```
<button type="submit">
  Send enquiry
</button>
<button type="button">
  Show help
</button>
```

TAKEAWAY **Use clear action labels such as “Send enquiry”.**

GET versus POST

Two common methods for sending form values.

GET

Values commonly appear in the URL query string; useful for public searches.

POST

Values are sent in the request body; often used for requests.

Security

POST alone is not encryption: HTTPS and access controls still matter.

TAKEAWAY **Never place private record details in public URLs.**

Why HTML alone cannot store requests

An institutional service includes more than the form.

Interface

HTML shows fields and gathers input in the browser.

Processing

A suitable server or service receives and validates requests.

Data protection

Storage, permissions and audit requirements need enforcement.

TAKEAWAY **A visible Submit button is not proof of a working system.**

Book-request form example

A library form can capture a clear and limited set of details.

Title

The visitor identifies the book to request.

Email

A contact channel is supplied if needed.

Collection point

One location is chosen using radio buttons.

HTML EXAMPLE

```
<label for="book">Book</label>
<input id="book" name="book"
       required>

<label for="email">Email</label>
<input id="email" name="email"
       type="email">
```

TAKEAWAY **The back end must check availability before confirming a request.**

Explain the library form

Think about what each control contributes.

Required fields

Ask for a book title and contact details only where justified.

Choice

A group of radio buttons can capture a collection preference.

Feedback

A real service should confirm receipt or explain errors.

TAKEAWAY **Design forms around real service tasks, not unnecessary questions.**

Public records enquiry

A short enquiry can identify a public record category.

Reference

Use a fictional demonstration reference.

Category

Select a published report or service guide.

Question

Use a textarea for the visitor’s enquiry.

HTML EXAMPLE

```
<label for="ref">Reference</label>
<input id="ref" name="ref"
       required>
<label for="cat">Category</label>
<select id="cat" name="category">
  <option value="reports">
    Public reports</option>
</select>
```

TAKEAWAY **Public enquiries must respect records access policies.**

Privacy and access control

The design of a form should support responsible information handling.

Data minimisation

Request only what the task requires.

Access rules

Never assume that a visitor may retrieve a restricted file.

Safe practice

Use fictional references and sample contact details in exercises.

TAKEAWAY **Do not publish confidential records in a demonstration website.**

Live editing: start with a form

Change the form and observe its visible behaviour.

Step 1

Open the library or records example.

Step 2

Change a label and add required to one suitable input.

Step 3

Preview the form; compare fields and test an empty response.

TAKEAWAY **The preview demonstrates controls, not real request processing.**

Common form mistakes

Small markup choices can make a form frustrating.

Missing labels

Visitors may not know what an empty box means.

Wrong control

Multiple checkboxes should not replace a single-choice group.

False promises

Do not claim that a static practice form has saved a request.

TAKEAWAY **Clarity, accessibility and truthful feedback build trust.**

Create one simple form

A manageable independent HTML exercise.

Two fields

Use meaningful labels and at least one required input.

One choice

Add radio buttons, a checkbox or a dropdown.

Evidence

Submit an HTML file and one screenshot of the form.

TAKEAWAY **A database or JavaScript is not required for this assignment.**

What you should remember

HTML forms support useful interaction with webpages.

Structure

form, label, input, textarea, select and button.

Experience

Choose controls carefully and include browser validation.

Responsibility

Processing, storage and permissions require more than HTML.

TAKEAWAY **Next: CSS will make the forms and webpages easier to read.**